EXOSCOPE

A terminal-based, retro-futuristic exoplanet exploration simulator. Analyze telemetry data, decrypt alien signals, and explore the cosmos through a CLI-based HUD. Built using Lazarus/FPC.



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CHAPTER 1:
FACILITY INTRODUCTION (INTRODUCTION)

Welcome to EXO-SCOPE Research Command

Congratulations on your new assignment in the Deep Space Telemetry Division. You now have full control of EXO-SCOPE, the most advanced exoplanet monitoring satellite network ever created by humanity.

As a Telemetry Operator, you won't be flying a spaceship. Your battlefield is a sea of data, spectral waves, and light curve graphs.

Your role Your daily tasks include:

1. Directing long-range satellites to scan unexplored star sectors in the galaxy.
2. Removes cosmic signal noise from raw data transmissions using advanced filtering techniques.
3. Identifying the presence of exoplanets based on transit anomalies on starlight charts.
4. Analyzing spectrum waves to detect the presence of water, oxygen (Biosignature), or more surprisingly... giant metal artificial structures left behind by alien civilizations.

Your Main Goal Space operations are incredibly expensive. This facility thrives on your inventions.

- **Research Funding:**Each validated planet discovery will earn you funding from Earth's government. Planets in the "Habitable Zone" will provide significantly more funding, and the discovery of alien anomalies will make you an instant trillionaire. Use these funds to launch Surface Probes or upgrade equipment.
- **Discovery Points:**Your reputation as an astronomer. The more celestial objects you catalog, the higher your points.
- **Survive:**The universe is dark and full of surprises. Beware of empty sectors with extreme gravitational anomalies. Directing a satellite into a black hole will destroy hundreds of thousands of dollars worth of equipment and cut off your research funding!

You are Earth's eyes and ears in the darkness of the cosmos. Don't miss a single signal.

CHAPTER 2:
DASHBOARD INTERFACE (UI OVERVIEW)

EXO-SCOPE Main System: Spectrum Ops

Before you lies the EXO-SCOPE operational dashboard. This interface is designed for maximum efficiency. As an operator, you don't use a joystick or manual steering; your primary tools are command lines and data manipulation.

Here is the layout of your workstation:

1. Command Terminal (Right Panel) This is the heart of your operation. On this black background panel (`USR@EXOSCOPE_OPS >`), you'll type all the satellite's instructions, from pointing the telescope at a new star sector, scanning for life signs, to triggering the launch of a surface satellite. This terminal system connects directly to the satellite in deep orbit with zero latency.

2. Chart Analysis Panel (Top Left) This area displays the raw data sent by the satellite, divided into two main instruments:

- **Light Curve Analysis (Main Chart):**Shows fluctuations in the relative flux (star brightness) over time. If a planet passes in front of the star (transit), this curve will exhibit periodic dips. The initial data is usually very dirty and full of random white dots (noise).

- **Spectrum Chart (Bottom Graph):**A small graph that maps the wavelengths of light (in nanometers/nm). This graph is crucial for identifying specific materials or artificial anomalies in a target's atmosphere.

3. Control Dock & Parameters (Bottom Panel)A manual control center for cleaning and processing cosmic signals. You can manipulate parameters using colored sliders before executing:

- **Filter Type & Threshold (Red Slider):**Select a cleaning algorithm (such as GAUSSIAN or MOVINGAVG) and set its sensitivity threshold.

- **Bandwidth (Green Slider) & Noise Gate (Blue Slider):**Sets how much of the background spectrum waves are filtered out.

- **Blue "Apply Filter" Button:**Runs a cleaning algorithm on the Light Curve graph based on the parameters you set.

- **Green "Report" Button:**Submit the final, clean analysis to Earth Command Center.

4. Central Visualizer (Bottom Center) This holographic projection screen has two main functions:

- **Vector Visualizer:**Displays a real-time 3D orbital projection of the solar system being monitored, complete with central star, planetary orbital wires, a scattering of faint stars, and a haze of cosmic nebulae.

- **Galaxy Map (Starmap):**If you use the command `VIEW -MAP`, this screen will expand to show thousands of star sectors in the galaxy ready to be explored.

5. Status Bar (Top Header) Never ignore this line. Here you can monitor the most vital operational indicators:

- **Research Funding:**Your current fund balance (e.g., $1,200,000). If this number runs out, your mission fails.

- **Discovery Points:**Your reputation and exploration score.

- **Equipment Status:**Shows your current technology level (Sensor Lv. 1 & Radar Lv. 1).

CHAPTER 3:
STANDARD OPERATING PROCEDURE (CORE GAMEPLAY LOOP)

As an EXO-SCOPE operator, your daily routine revolves around three key steps: Acquisition, Processing, and Reporting. Strictly follow these protocols to ensure the uninterrupted flow of research funding to our facility.

STAGE 1: Target Acquisition (Finding Stars) Your first step is to find a star to observe. You can't analyze the void.

1. Type the command `VIEW -MAP` in Terminal to open the Galaxy Navigation Map. Your central visualizer screen will change to display a projection of the galaxy.
2. In the Galaxy Map, you will see hundreds of points of light representing sectors of the solar system.
3. You can click on one of the star points using the mouse, or if you know the exact coordinates, type the command directly.`CONNECT -SECTOR <SECTOR_NAME>` (Example:`CONNECT -SECTOR HD-9362`).
4. The satellite will automatically lock onto the target, download the raw data, and move you back to the chart analysis dashboard (`VIEW -CHARTS`
(Warning: Be careful when selecting sectors. Read Chapter 5 on gravity anomalies before you recklessly connect to any suspicious-looking area.)

STAGE 2: Signal Processing (Data Cleaning) The newly downloaded starlight data is highly susceptible to cosmic radiation interference (noise). You'll notice your Light Curve graph is littered with random, messy dots. Your task is to clean them up.

1. Look at the Light Curve chart. Your goal is to look for a "Transit Curve"—a U-shaped dip in the light that occurs periodically. This dip indicates a planet is passing through the star's light.
2. Since noise covers the depression, you need to filter it. Set the filter type in the Control Dock (for example: `GAUSSIAN`), then adjust the Threshold using the red slider.
3. Click the blue "Apply Filter" button in the bottom panel, or type the command `FILTER - GAUSSIAN MODE`at the terminal.
4. Observe the chart. If the U-shaped curve is clearly visible and the pattern is stable, you're ready to proceed with reporting. If not, adjust the parameters and filters again.

STEP 3: Expedition Reporting (Funding Claim) Once you are sure you have found a planet (indicated by a clear transit curve), you must summarize and send the report to Earth.

1. Click the green "Report" button in the bottom right corner of the Control Dock, or type the command `REPORT` at the terminal.
2. The system will automatically calculate the number of planets, their radius and position.
3. **Funding:**If your analysis is correct and the system detects the presence of a planet, you'll receive a Research Funding injection. The value will skyrocket if the planet is in the Habitable Zone.
4. **Penalty:**Never report "Empty Stars" (charts that lack planetary basins). If you press REPORT on a star with no planets, the Command Center will consider you wasting the satellite's time, and your research funding will be deducted as a penalty.

CHAPTER 4:
ADVANCED OPERATIONS & SPECIAL EXPLORATION (ADVANCED FEATURES)

Detecting the presence of a dead rock or a giant ball of gas orbiting a star is a standard scientific achievement. However, the primary reason for this facility's existence is to search for the "Needle in the Cosmic Haystack": Organic Life and Extraterrestrial Technology.

While you are analyzing a sector (in Data Acquired status), you can use the following experimental features via the Command Terminal:

1. Spectrum Anomaly Scan (`SCAN -BIO`) If you see the Spectrum Chart graph in the bottom left displaying an unusual spike (e.g. a green line indicating O2/H2O, or fuchsia indicating artificial Titanium/Iron), perform an auto scan immediately.

- **Order:**Type `SCAN -BIO`at the terminal.
- **Effect:**The system will perform a deep spectroscopy. With luck, the terminal will confirm the presence of a biosignature (a living world) or an alien megastructure (a Dyson Swarm/Synthetic Structure).
- **Profit:**Don't forget to do `REPORT` After this confirmation! The discovery of an anomaly will multiply your Research Funding massively (up to millions of dollars).

2. SETI Radio Receiver (`LISTEN -FREQ`) Turn off your visual instruments for a moment and switch to audio. The integrated Audio Synthesizer module lets you "listen" to space signals in real time.

- **Order:**Type `LISTEN -FREQ <NUMBER>` (Example:`LISTEN -FREQ 1420`).
- **Secret Frequency:**
- `1420` **MHz (Hydrogen Line):**This universal frequency is often used to search for civilizations. If there's a pattern of organic life, you'll hear the rhythmic beat of the data (biosignature).
- `1618` **MHz (Golden Ratio):**High-energy distortion frequencies. If you hear a heavy, synthetic drone, you're dealing with a Megastructure.
- To turn off the radio, simply type `LISTEN` without argument.

3. Surface Imaging Satellite (`LAUNCH -PROBE`) Graphs and numbers are scientific, but humans need to see the real thing. If you have ample research funding, you can launch a Surface Probe satellite to photograph the target surface.

- **Order:**Type `LAUNCH -PROBE` at the terminal.

- **Cost:**Be careful, this operation is very expensive! It requires $1,000,000 of your Research Funding for a single launch.
- **Results:**The system will render a high-resolution 2D photo (procedural pixel art) of the planet's landmass or alien structure. This valuable photo will be automatically saved in your computer's directory (in the folder`PROBES/`) as an exploration trophy!

4. Official Report Documentation (`EXPORT -PDF` **)**As an authorized operator, you are required to archive important discoveries into the Earth database archive.

- **Order:**Type `EXPORT -PDF` when you finish analyzing an inhabited sector.
- **Results:**The system will generate a physical, printed document in PDF format (Classified Discovery Report). This report includes the star's classification, number of planets, Habitable Zone status, and a secret stamp. This file will be saved in the folder `REPORTS/`on your local computer system.

5. Anomaly Scanning & Emergency Alerts (`SCAN -BIO`/`LISTEN`**)** If you suspect an anomaly in a sector, you can use the command`SCAN -BIO`(for visual scanning) or `LISTEN -FREQ 1420`/ **`1618`** (for audio detection).

- If the sector is empty, the system will report it as a Dead World.
- **HOWEVER, IF ALIENS ARE DETECTED:**Instead of immediately providing results, the satellite defense system will trigger an Emergency Protocol (Red Alert). Your visualizer screen will pulse red, and the terminal will display: "UNKNOWN ENCRYPTED TRANSMISSION DETECTED. SIGNAL DECAY IN 60 SECONDS."

6. First Contact Protocol: Signal Decryption (`DECODE` **)**When a Red Alert occurs, your normal Light Curve graph will be overwritten by the Signal Intercept interface. You will see two radio waves:

- **Target Wave (Fuchsia/White):**This is a very complex alien signal pattern.
- **Your Satellite Wave (Cyan):**This is your current receiver frequency.
- **Your mission:**You only have 60 seconds before the signal fades and the target data is completely corrupted. You must guess and align the Cyan Wave so that its shape overlaps exactly with the Fuchsia Wave.
- **Ways of working:**Type the command `DECODE -FREQ <NUMBER> -PHASE <NUMBER>` repeatedly to shift your wave.
- *FREQ (Frequency) Range:*`1.0` until `9.0` (Tightening or loosening the waves).
- *PHASE Range:* `0.0` until `9.9`(Slide the wave left/right).
- *Example:*`DECODE -FREQ 5.5 -PHASE 2.0`

7. Risk & Reward (High Stakes)

- **If Successful (Decrypted):**The key is unlocked! Visuals of alien exoplanets will be revealed, and Earth Command Center will be awarded a massive funding bonus of $10,000,000.
- **If Failed (Time Out):**The satellite connection is forcibly lost. The sector data is permanently corrupted, and you must search for targets in other stars from scratch without any pay.

8. Documentation & Imaging (`LAUNCH -PROBE` & `EXPORT -PDF`) After successful decryption and status becomes Reported:

- Use `LAUNCH -PROBE` ($1,000,000) to photograph the surface of the planet or Megastructure (images are stored in the folder `PROBES/`).
- Use `EXPORT -PDF`to print the "TOP SECRET" confidential document of your invention (stored in the folder `REPORTS/`).

CHAPTER 5:
COSMIC THREATS & ANOMALIES (HAZARDS & SECRETS)

Space isn't just about beautiful planets and stars. As an operator, you must possess a keen sense of visual acuity. The universe holds deadly traps and the remains of ancient civilizations waiting to be discovered.

1. Event Horizon (Black Hole & Gravitational Lens) Not all points on the Galactic Starmap are safe to explore. Some ghost sectors have collapsed into black holes.

- **How to Recognize:**On Starmap, black holes emit no light, so they appear as pitch-black dots or empty areas. However, look closely at the stars around them. If you notice their positions appear "warped" or distorted as they pass through the radar sweep, you're observing gravitational lensing. Their massive gravitational pull bends light!
- **FATAL DANGER:**If you recklessly (or accidentally) carry out the command `CONNECT -SECTOR` If you were to approach the center of the anomaly, satellites worth hundreds of thousands of dollars would be sucked into the Event Horizon and instantly destroyed. You would receive a critical red alert, lose telemetry connections, and worst of all, your $200,000 Research Fund would be instantly forfeited as damages.

2. Alien Megastructures (Dyson Swarm) This is the primary (and rarest) target of the EXO-SCOPE project. Sometimes, starlight fluctuations are caused not by natural planets, but by giant artificial structures built to harvest the star's energy (such as Dyson Swarms).

- **Visual Characteristics:**If you've successfully cleared the graph data, take a look at the Vector Visualizer screen. The central star will appear to be glitched, and its orbit will no longer be a typical planetary circle. You'll see a hexagonal or diamond-shaped geometric structure with a purple/fuchsia neon glow surrounding the star.
- **Spectral Characteristics:**Take a look at the Spectrum Chart. If there's a strange spike at a wavelength around 500nm (Fuchsia), it indicates the presence of a pure synthetic metal alloy (Titanium/Iron) on a planetary scale.
- **Action:**Use the command immediately `SCAN -BIO` And `LISTEN -FREQ 1618` to confirm your findings, launch `LAUNCH -PROBE` if you want a photo of the structure, and top it off with `REPORT` to claim billions of dollars in funds!

CHAPTER 6:
RESEARCH & SYSTEM IMPROVEMENT (TECH TREE & ECONOMY)

Space operations are not a charity. Every data transmission, satellite time lease, and probe launch costs a significant amount of money. As an operator, you act as both a scientist and a financial manager for your division.

1. Funding System (Research Funding) Your research funding is the lifeblood of this mission. You begin your career with a base grant of $100,000.

- **Main Income:**Obtained by executing commands `REPORT` on valid findings.
- *Ordinary Planets (Gas Giant / Terrestrial):*Standard funding ($50,000 - $500,000).
- *Habitable Zone:*Additional massive bonus ($500,000).
- *Biosignature (Sign of Life):*Extraordinary funding ($2,000,000).
- *Alien Megastructure:*Unlimited grant from the earth government ($5,000,000).

- **Expenses & Penalties:**
- False Report (Doing `REPORT` on a star without planets): Fine -$10,000.
- Black Hole Tragedy (Sucked into Event Horizon): Satellite repair fine -$200,000.
- Surface Probe Launch: Operating cost -$1,000,000.

2. Equipment Improvements (Upgrades) Use your surplus research funds to upgrade the EXO-SCOPE satellite hardware capabilities.

- `UPGRADE - SENSOR` **(Cost: $300,000)**Upgrade the optical lens to Level 2. This advanced sensor will drastically reduce the inherent cosmic noise in the raw data, making it easier and faster to find transit curves without the need for multiple filters.
- `UPGRADE - RADAR` **(Cost: $500,000)**Upgrade the Deep Space Radar array to Level 2. This system unlocks remote Star Classification. With this radar, when you connect to a sector, the system will instantly recognize its Star Type (such as O, B, A, F, G, K, M), Mass, and Temperature before you analyze it.

CHAPTER 7:
EMERGENCY PROTOCOL & TERMINAL REFERENCE (CHEAT SHEET)

When performing remote telemetry operations, you may encounter unexpected situations or technical difficulties. This final chapter includes a troubleshooting guide and a complete list of terminal commands for quick reference at your desk.

1. Troubleshooting Protocol

- **Case: My light curve graph is just a mess of random dots.**
- **Solution:**Raw data is always full of cosmic radiation (Noise). Make sure you use the command `FILTER - GAUSSIAN MODE` Or, drag the Threshold in the Control Dock and then press Apply Filter until the dots form a smoother line. If you've upgraded the Sensor to Level 2, the initial noise will be significantly reduced.

- **Case: I have filtered the data, but the graph line is completely straight without any dips.**
- **Solution:**This means you are monitoring a dead star that has no planets. Stop operation immediately and do not press `REPORT` (You will be fined!). Just type `VIEW -MAP` and look for other sectors.

- **Case: My Research Funding is approaching $0.**
- **Solution:**You're in danger of being fired from the project. Stop launching the Surface Probe ($1,000,000) and focus on finding planets in the Habitable Zone, or search for signals from the Alien Megastructure to secure a massive funding injection.

- **Case: Terminal displays message "CRITICAL WARNING: EVENT HORIZON DETECTED".**
- **Solution:**There's no solution. You've just directed a satellite into a black hole. Accept the $200,000 satellite damage penalty, return to the Galaxy Map, and avoid sectors where the surrounding starlight appears curved (gravitational lensing).

2. Terminal Quick Reference (Command Cheat Sheet)

Use this list of pure commands as a quick cheat sheet. Type these commands exactly as written (note the use of spaces and hyphens):

Navigation & Visuals

- `VIEW -MAP`: Switch to the Galaxy Navigation Map screen.
- `VIEW -CHARTS`: Switch back to the instrument and graph dashboard screen.
- `CLEAR`: Clears the history text from your terminal screen so it doesn't fill up.
- `HELP`: Displays a short list of commands directly on the terminal screen.

Operations & Analysis (Main Cycle)

- `CONNECT -SECTOR <NAME>`: Locking the satellite target to a specific star sector (Example :`CONNECT -SECTOR HD-9362`).
- `FILTER -MODE <TYPE>`: Implements a light data cleaning algorithm. Available types: `GAUSSIAN` or `MOVINGAVG`. Additional parameters such as`-SIGMA`automatically set by the slider in the Control Dock.
- `REPORT`: Analyze the cleared curve, claim the discovery of the planet, and disburse Research Funds from Earth.

Advanced Exploration

- `SCAN -BIO`: Perform deep spectroscopic scans to confirm signs of life (O2/H2O) or traces of alien synthetic metals.
- `LISTEN -FREQ <MHZ>`: Tuning the cosmic receiver radio module.
- *Tip:*Use `1420` to detect Biosignature or `1618` for Megastructure.
- *Tip:*Type `LISTEN` without frequency parameters to mute the audio.

- `LAUNCH -PROBE`: Launch a photo satellite to the target surface to obtain 2D (pixel-art) images. Cost: $1,000,000.
- `EXPORT -PDF`: Prints a secret PDF document archive of the active solar system.

Procurement & Research (Economics)

- `UPGRADE - SENSOR`: Purchase an optical sensor module upgrade to Level 2 (Cost: $300,000).
- `UPGRADE - RADAR`: Purchase the Deep Space Radar module upgrade to Level 2 to unlock detailed star classification on the map (Cost: $500,000).

3. Additional Troubleshooting Protocols

- **Case: The screen flashes red, there is a "DECRYPTION IN PROGRESS" warning, and my Light Curve graph turns into a weird wavy line.**
- **Solution:**DON'T PANIC! You've just detected an alien signal, but it's encrypted. Use the command immediately.`DECODE -FREQ <VALUE> -PHASE <VALUE>` to match your Cyan waves to the aliens' Fuchsia waves. You only have 60 seconds before the signal disappears. The Filter and Report buttons will be disabled during this emergency.

2. Terminal Quick Reference (Command Cheat Sheet)

- `HELP`: Displays a short list of commands in the terminal.
- `VIEW -MAP`: Opens the Galaxy Navigation Map.
- `VIEW -CHARTS`: Back to the instrument and chart dashboard.
- `CONNECT -SECTOR <NAME>`: Locking the satellite target to a specific sector (Example: `CONNECT -SECTOR HD-9362`).
- `FILTER -MODE <TYPE>`: Applies noise cleaning. Available types: `GAUSSIAN` or `MOVINGAVG`.
- `SCAN -BIO`: Perform spectroscopic scans to search for life/anomalies.
- `LISTEN -FREQ <MHZ>`: Tune in to the cosmic listening radio (Use `1420` or `1618`). Type `LISTEN` without argument to turn off.
- `LAUNCH -PROBE`: Launch a photo satellite to the planet's surface ($1,000,000).
- `EXPORT -PDF`: Prints an official PDF document archive of the solar system being analyzed.
- `REPORT`: Claiming inventions and getting funding.
- `UPGRADE - SENSOR`: Purchase Sensor module upgrade ($300,000).
- `UPGRADE - RADAR`: Purchase Radar module upgrade ($500,000).
- `CLEAR`: Clears text from the terminal screen.

Special Exploration & Decryption

- `SCAN -BIO`: Performing a deep spectroscopic scan. (May trigger a Decryption Emergency if aliens are detected).
- `LISTEN -FREQ <MHZ>`: Tune in to the cosmic listening radio (Use `1420` or `1618`). Type `LISTEN` without argument to turn off. (Can trigger Emergency Decryption state).
- `DECODE -FREQ <1.0-9.0> -PHASE <0.0-9.9>`: Guess and shift satellite radio waves to match alien signal codes. CAN ONLY BE USED when the screen is flashing red (Red Alert).
- `LAUNCH -PROBE`: Launch a photo satellite to the target surface ($1,000,000).
- `EXPORT -PDF`: Print confidential PDF document archives.